

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of M. Tech. (IT) Examination

December 2013

IT 708 Software Architectures & Frameworks

Date: 24.12.2013, Tuesday

Time: 10:00 a.m. To 1:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) What is Design Pattern? Explain all types of design patterns with atleast five [07]
examples of each.

Q - 2 (a) What is MVC? Explain MVC Design Pattern. (Include Example with code snippet) [07]

(b) Explain Command Design pattern with Example and its UML and Code Snippet. [07]

OR

Q - 2 (a) Explain Display Design pattern with Example and its UML and Code Snippet. [07]

(b) Explain Chain of Responsibility Design pattern with Example and its UML and Code [07]
Snippet. Also state problems in chain of responsibility design pattern.

Q - 3 (a) Explain following terms with respect to AOP. [03]

1) Pointcut 2) Dataflow Relation 3) Excluding Condition

(b) Differentiate: OOP v/s AOP [04]

(c) State how reader writer problem can be implemented using Aspect oriented [07]
programming. Also explain problems with OOP solution.

OR

Q - 3 (a) Draw following diagrams for ATM system. [07]

1. Join Point Identification diagram

2. Behavioral Join Point Model for Cash Withdraw

3. Aspect Design Diagram

4. Aspect-Class Dynamic Model for Cash Withdraw

(b) Draw design notations for AODL for following components. [07]

1. Join Point 2. Pointcut

3. Aspect 4. Code Weaving

Also explain each term in detail with example.

SECTION – II

- Q - 4 (a) What is framework? Explain feature of framework in detail. [05]
(b) What is Hibernate? Explain Hibernate Sessions. [02]
- Q - 5 (a) What is spring framework? Explain with architecture. [07]
(b) Explain BeanFactory interface and their methods. [07]

OR

- Q - 5 (a) What is the difference between java4 and java5? Explain following features of java5. [07]
1. Autoboxing
2. Scanner Class
3. Generics
4. Iterator
5. Typesafe Enum
- (b) What is POJO? How to use it? Explain it with example. [07]
- Q - 6 (a) What is dependency injection? Explain their types with example. [07]
(b) How to access application components and resources? [07]
